Touchpoint Constellation

An Explainable, Interactive Visual Analytics System for Multi-Touch Marketing Attribution

Course: Advanced Data Visualization Techniques
Domain: Business / Marketing Analytics
Project type: Visualization-first system with an ML engine

1. One-Line Pitch

Same data, different measurement method, different business decision. Touchpoint Constellation turns customer journeys into a living star map so a marketer can watch channel importance change as the attribution model changes, and see exactly where budget is being misallocated.

2. The Problem

Most businesses decide their marketing budget using **last-click attribution**, which gives 100% of the credit for a sale to the final touchpoint a customer used. This ignores every channel that built awareness and interest earlier in the journey.

The consequences are real:

- Upper-funnel channels such as Display and Social look useless on reports, so their budgets get cut.
- Bottom-funnel channels such as Search and Email look like heroes, although they often only collect demand created elsewhere.
- Marketers cannot see which channel *combinations* work together, because standard dashboards show channels one at a time.
- Attribution reports are tables of numbers. Decision-makers cannot *feel* how much the choice of model changes the answer.

3. The Proposed Solution

An interactive web application that models customer journeys with machine learning and presents the results as a **constellation**:

- Every marketing channel is a **star**.
- Every customer journey is a **glowing arc** linking the stars the customer touched before converting.

- A single toggle switches the attribution model, and the stars **physically grow and shrink** with smooth animation.

Every visual element is the output of a model, so nothing on screen is decoration.

4. Novelty and How to Explain It

The novelty has three parts, each explainable in one sentence:

1. **A spatial metaphor instead of a bar chart.** Attribution is usually shown as tables and bars. A star map lets users see channel influence and channel relationships at the same time.
2. **Live model comparison on identical data.** Switching between Last-Click, Linear, Markov, and Shapley attribution makes the *cost of measurement choice* visible in a way static reports cannot.
3. **Explainable and uncertainty-aware.** Each journey can be explained with SHAP, and every credit estimate carries a bootstrap confidence halo, so users see how much to trust it.

30-second explanation for the audience:

> Marketers allocate budgets using last-click attribution, which ignores everything that built demand before the final click. My visualization turns customer journeys into a constellation, so you can watch channel importance change live as you switch attribution models, and see exactly where money is being misallocated. It is not a new chart for numbers. It is a new way to *see* how the measurement method changes the decision.

5. Visualization Design

Four linked views. Clicking a star filters all the others.

View	What it shows	Visualization technique
A. Constellation (hero view)	Channels as stars, journeys as arcs, star size changes with the selected model	Force-directed layout, arc diagram, animated transitions
B. Truth Gap panel	Credit per channel under Last-Click vs. Shapley, with the biggest misallocations highlighted	Dumbbell chart

View	What it shows	Visualization technique
C. Journey Replay	A comet travels one customer's path star to star, with a live conversion probability and a SHAP explanation	Animated path, line chart, diverging bars
D. Budget Simulator	Drag spend between channels and see the estimated change in conversions with uncertainty	Interactive sliders, fan chart, uncertainty halos

Visual encoding

Visual element	Data it encodes
Star size	Attributed credit under the selected model
Star glow	Predicted conversion contribution (from the XGBoost model)
Star halo width	Bootstrap uncertainty of the credit estimate
Arc thickness	How often two channels appear together in converting journeys
Arc color	Journey cluster (customer type)

The design rule: **everything that glows, moves, or changes size must encode data.**

The five "wow" moments for the demo

1. **The Model Switch.** Toggle Last-Click to Shapley and watch the stars reshuffle.
2. **The Truth Gap.** A single number showing how much budget is misallocated under last-click.
3. **Journey Replay.** A comet traces one customer's path while the conversion probability rises and falls.
4. **Synergy Pairs.** Hover an arc to see how much more often two channels convert together than alone.
5. **Budget Reallocation.** Drag budget between stars and watch the estimated conversions change.

6. Machine Learning Component

Prediction task

Binary classification at the journey level: given an ordered sequence of touchpoints, predict the probability that the journey ends in a conversion.

Item	Definition
Unit of data	One customer journey (ordered list of touchpoints)
Target	1 if the journey converted, 0 otherwise
Baseline model	Logistic regression
Main model	XGBoost / LightGBM
Features	Channel counts, first and last channel, path length, time span, average gap between touches, recency, channel transitions, click vs. view flags where available
Evaluation	Time-based or user-based split, class-imbalance handling, ROC-AUC, PR-AUC, calibration plot

ML components and where they appear on screen

ML technique	Purpose	Visual output
Gradient boosting	Predict conversion probability	Star glow, replay probability, simulator forecast
Shapley value attribution (sampled)	Fair channel credit using game theory	Star size
Markov chain removal effect	Credit based on how much conversion drops if a channel is removed	Star size, model comparison
SHAP	Explain why a specific journey was predicted to convert	Replay panel bars
k-means clustering	Discover customer journey types	Arc colors, small multiples
Bootstrap resampling	Quantify uncertainty in credit estimates	Halos around stars

Important distinction: attribution models distribute credit for conversions that already happened. The XGBoost model predicts the probability of conversion. They serve different parts of the interface.

7. Data

Dataset	Role	Notes
Google Analytics sample (Google Merchandise Store)	Primary candidate	Public BigQuery dataset with named channels (Organic Search, Direct, Social, Paid Search, Referral, Display, Affiliates). Availability and conversion volume to be verified before building.
Criteo Attribution Modeling for Bidding dataset	Validation dataset	Free from Criteo AI Lab, about 16M impressions over 30 days, campaign-level rather than named channels, requires accepting Criteo's terms of use.

Dataset	Role	Notes
Synthetic journeys with known channel effects	Correctness test	Lets us prove that Shapley and Markov recover true channel value while last-click does not. Clearly labeled as synthetic.

8. System Architecture

```
Raw data (GA sample / Criteo / synthetic)
        |
   Journey builder (ordered touchpoints + conversion flag)
        |
   ML engine (Python): XGBoost, Shapley, Markov, SHAP, k-means, bootstrap
        |
   Aggregated JSON files
        |
   D3.js front end: four linked views
```

Technology stack: Python (pandas, NumPy, scikit-learn, XGBoost, SHAP) for the pipeline, JSON as the data contract, D3.js with SVG and canvas for the visualization, hosted as a static web page.

9. Evaluation Plan

- **Prediction quality:** logistic regression vs. XGBoost using ROC-AUC, PR-AUC, and a calibration plot.
- **Attribution validity:** on the synthetic dataset, compare each model's credit against the known ground truth.
- **Stability:** report bootstrap interval widths per channel.
- **Usability:** a short walkthrough with 3 to 5 classmates, asking them to find the most misallocated channel using the tool.

10. Three-Day Plan

Day	Focus	Deliverables
Day 1	Data and ML	Clean journeys, rule-based and Markov attribution, baseline and XGBoost models, JSON export

Day	Focus	Deliverables
Day 2	Visualization	Constellation with model switch, Truth Gap panel, linked filtering; Shapley, SHAP, and bootstrap added to the pipeline
Day 3	Integration and polish	Journey Replay, Budget Simulator, clustering colors, dark theme, story mode, report, demo video

Fallback if time runs short: the constellation, model switch, and Truth Gap alone already deliver the core idea. The replay and simulator are the first features to defer.

11. Deliverables

- Interactive web application (D3.js)
- Python notebook with the full ML pipeline
- Short report covering the problem, method, visual encoding table, and results
- Three-minute demo video as a backup for the live demo

12. Risks and Mitigation

Risk	Mitigation
Shapley is computationally heavy	Use sampled approximation and state this openly
Dataset has too few conversions or unlabeled channels	Fall back to the second dataset or to the synthetic data with known effects
Visual effects consume too much time	Build a plain working version first, then add glow and animation last
Simulator results are mistaken for causal claims	Word results as "estimated change under the model", not guaranteed gains
Scope creep	Keep to 5 to 6 channels and follow the defer list above

13. Visualization Techniques Demonstrated

Force-directed layout, arc and chord diagrams, animated transitions, dumbbell charts, Sankey path flows, small multiples, uncertainty visualization (halos and fan charts), linked brushing and filtering, diverging bar charts for SHAP, calibration and ROC plots, and narrative storytelling with a guided story mode.

14. Expected Outcome

A polished, explainable analytics tool that shows, on real or realistic data, that the choice of attribution model can change which channels look valuable, and gives a marketer a clear, visual basis for reallocating budget. The project demonstrates advanced visualization design and a rigorous, honest use of machine learning, where every model output has a visible purpose on screen.